

CIFAR-10 $f^*(2\sigma)$ tables: pre- and post-discovery, both cuts

Every value is $f^*(2\sigma)$ — the smallest injected fraction detected at 2σ median expected significance (lower is better) — median over draws h4/7/8/9. Residual children indented. Best (lowest) test per row in **bold**.

method	Eucl.	Maha.	MMD	SparKer
SupCon	>.1	.0569	.0199	.0176
SupCon+SIGReg	>.1	>.1	.0308	.0368
NPLM-dist	>.1	.0383	.0394	.0268
NPLM+cw-SIGReg	>.1	.0318	.0325	.0266
proto/MSE+cw-SIGReg	>.1	.0592	.0413	.0388
SupCon+cw-SIGReg	>.1	.0480	.0334	.0579
SupCon → res	>.1	.0225	.0331	.0222
SupCon → res-nplm	>.1	.0230	.0294	.0171
SupCon → res	>.1	.0290	.0442	.0689
SupCon → res-nplm	>.1	.0439	.0380	.0219
NPLM-dist → res	>.1	.0412	.0464	.0457
NPLM-dist → res-nplm	>.1	.0484	.0442	.0390
NPLM → res	>.1	.0226	.0371	.0478
NPLM → res-nplm	>.1	.0339	.0361	.0425
proto/MSE → res	>.1	.0615	.0442	.0388
proto/MSE → res-nplm	>.1	.0529	.0431	.0428

Table 1: CIFAR-10 pre-discovery (frozen) $f^*(2\sigma)$, median over 4 draws. Cut-independent. Euclidean is censored on every arm — raw distance-to-centroid is a dead frozen novelty test.

method	Eucl.	Eucl.(a)	Maha.	MMD	SparKer	SpK(a)
SupCon	.0141	.0127	.0158	.0250	.0168	.0142
SupCon+SIGReg	.0373	.0400	.0636	.0343	.0367	.0425
NPLM-dist	.0446	.0207	.0369	.0358	.0322	.0247
NPLM+cw-SIGReg	.0436	.0154	.0310	.0444	.0327	.0203
proto/MSE+cw-SIGReg	.0374	.0500	.0425	.0610	.0374	.0650
SupCon+cw-SIGReg	.0406	.0300	.0487	.0351	.0475	.0500
SupCon → res	.0168	.0146	.0157	.0303	.0169	.0183
SupCon → res-nplm	.0210	.0176	.0179	.0287	.0193	.0189
SupCon → res	.0528	.0300	.0300	.0417	.0506	.0463
SupCon → res-nplm	.0386	.0167	.0314	.0325	.0259	.0192
NPLM-dist → res	.0359	.0300	.0284	.0400	.0370	.0322
NPLM-dist → res-nplm	.0857	.0331	.0581	.0504	.0269	.0445
NPLM → res	.0398	.0400	.0230	.0440	.0327	.0400
NPLM → res-nplm	.0709	.0400	.0267	.0434	.0424	.0400
proto/MSE → res	.0297	.0200	.0223	.0457	.0387	.0259
proto/MSE → res-nplm	.0284	.0235	.0259	.0455	.0286	.0290

Table 2: CIFAR-10 post-discovery f^* , derived label-free cut (np pool), median over draws. Eucl.(a)/SpK(a) are the anchor-aware tests. Blank median where the arm declines on most draws.

method	Eucl.	Eucl.(a)	Maha.	MMD	SparKer	SpK(a)
SupCon	.0219	.0150	.0195	.0212	.0174	.0155
SupCon+SIGReg	.0257	.0228	.0301	.0350	.0244	.0221
NPLM-dist	.0293	.0193	.0248	.0415	.0276	.0344
NPLM+cw-SIGReg	.0222	.0141	.0223	.0477	.0249	.0201
proto/MSE+cw-SIGReg	.0276	.0251	.0394	.0349	.0378	.0354
SupCon+cw-SIGReg	.0310	.0249	.0265	.0325	.0366	.0423
SupCon → res	.0175	.0169	.0159	.0312	.0178	.0179
SupCon → res-nplm	.0207	.0164	.0183	.0243	.0209	.0190
SupCon → res	.0216	.0181	.0155	.0385	.0331	.0394
SupCon → res-nplm	.0174	.0156	.0224	.0289	.0210	.0168
NPLM-dist → res	.0202	.0156	.0184	.0458	.0256	.0266
NPLM-dist → res-nplm	.0532	.0177	.0461	.0400	.0427	.0415
NPLM → res	.0191	.0166	.0162	.0402	.0256	.0256
NPLM → res-nplm	.0225	.0153	.0193	.0329	.0306	.0291
proto/MSE → res	.0200	.0223	.0211	.0514	.0264	.0238
proto/MSE → res-nplm	.0283	.0247	.0267	.0415	.0277	.0243

Table 3: CIFAR-10 post-discovery f^* , $s/\sqrt{b}$ cut (np pool + Punzi), median over draws. Engages everywhere (fewer censored cells than the derived cut); the anchor-aware Euclidean on the SupCon family is best.